Coral Reefs (2014) 33:625–636
DOI 10.1007/s00338-014-1173-y

REPORT

Fish corallivory on a pocilloporid reef and experimental coral responses to predation

M. M. Palacios · C. G. Muñoz · F. A. Zapata

Received: 12 October 2013 / Accepted: 18 May 2014 / Published online: 1 June 2014
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Abstract This study examined the effects of the Guineafowl pufferfish (*Arothron meleagris*), a major corallivore in the Eastern Pacific, on pocilloporid corals on a reef at Gorgona Island, Colombia. Pufferfish occurred at a density of 171.2 individuals ha^{-1} and fed at a rate of 1.8 bites min^{-1}, which produced a standing bite density of 366.2 bites m^{-2}. We estimate that approximately 15.6 % of the annual pocilloporid carbonate production is removed by the pufferfish population. Examination of the predation effect on individual pocilloporid colonies revealed that although nubbins exposed to corallivory had lower linear growth, they gained similar weight and became thicker than those protected from it. Additionally, colonies with simulated predation injuries (on up to 75 % of branch tips) healed successfully and maintained growth rates similar to those of uninjured colonies. Despite the high corallivore pressure exerted by pufferfish on this reef, we conclude that they have a low destructive impact on *Pocillopora* colonies as corals can maintain their carbonate production rate while effectively recovering from partial predation. Due to its influence on colony morphology, pufferfish predation may increase environmentally induced morphological variability in *Pocillopora*.

Keywords *Arothron meleagris* · Corallivory · Reef fishes · Guineafowl pufferfish · *Pocillopora* spp. · Tropical Eastern Pacific

Communicated by Biology Editor Dr. Hugh Sweatman

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Introduction

Coral-feeding fishes, represented by at least 128 species (Cole et al. 2008), are an important functional guild within tropical coral reefs as they influence the abundance and distribution of coral species and coral community structure (Neudecker 1979; Wellington 1982; Hixon 1997; Rotjan and Lewis 2008; Lenihan et al. 2011). Most corallivore fishes are butterflyfishes (Chaetodontidae), which only remove live tissue from corals and cause little or no damage to skeletal structures (Cole et al. 2008). However, some pufferfish, parrotfish, and triggerfish scrape, excavate, or break corals removing also portions of the underlying calcareous skeleton. These fishes can have a significant impact on corals because the fitness of injured corals is usually altered through the reallocation of energy resources from growth and reproduction to tissue and skeletal regeneration (Hall 2001; Henry and Hart 2005). Such impact also contributes to determine the physical structure, zonation, and diversity of reef habitats (Wellington 1982; Cole et al. 2008; Jayewardene et al. 2009; Bonaldo and Bellwood 2011).

Studies on fish corallivory have examined the pressure and effects of predators on corals at different levels of organization. Some studies have determined the extent of damage inflicted or tissue removed by corallivores (Reyes-Nivia et al. 2004; Bonaldo and Bellwood 2011; Cole et al. 2011; Bonaldo et al. 2012;) and their effects on the growth or recovery of individual coral colonies (Jayewardene et al. 2009; Edmunds and Lenihan 2010; Lenihan and Edmunds 2010; Shantz et al. 2011). Others have examined the broader impacts of fish predation on coral species composition and coral community diversity (Cox 1986; Littler et al. 1989; Rotjan and Lewis 2006; Lenihan et al. 2011). While studies in the Caribbean and Indo-Pacific have focused on butterflyfishes and parrotfishes as major coral predators, those in the Tropical Eastern Pacific (TEP) have concentrated on the Guineafowl pufferfish *Arothron meleagris*—a generalist corallivore that ingests between 16 and 20 g $CaCO_3$ m^{-2} d^{-1} by constantly scraping and breaking off the branch tips of the abundant pocilloporid colonies (Glynn et al. 1972, 1982; Guzmán and López 1991; Reyes-Bonilla and Calderon-Aguilera 1999). This pufferfish, despite being perhaps the most important fish corallivore in the region (Guzmán and Robertson 1989), appears to have no significant negative impact on coral abundance, reef development, or community structure of pocilloporid coral reefs (Guzmán and López 1991; Reyes-Bonilla and Calderon-Aguilera 1999). It is unknown why the effect of puffers in the TEP differs from the common negative impact that corallivorous fishes exert on other coral reef regions (e.g., Rotjan et al. 2006; Bonaldo and Bellwood 2011; Shantz et al. 2011). Addressing this issue requires an understanding of the effects of fish predation on the growth and recovery of individual *Pocillopora* spp. colonies as well as basic information on the predator's feeding behavior, feeding rate, and the resulting predation pressure on the reef.

Considering that pocilloporid reefs in the TEP region differ greatly in species richness and composition from those in the Caribbean and Indo-Pacific, and that the pressure and effects of corallivory vary according to the characteristics of the fish predator and the coral prey (Rotjan and Lewis 2008; Lenihan and Edmunds 2010), conclusions from one system in a particular region cannot be easily extended to different systems in other regions. Before any generalizations can be made, specific studies must be undertaken. Additionally, as pocilloporid reefs in the TEP offer a simplified predator–prey system (only one important fish predator—*A. meleagris* and one main coral prey—*Pocillopora* spp.), it may be possible to have a better understanding of the interaction and the process of corallivory. This study examined corallivory by the puffer *A. meleagris* on pocilloporid reefs from the TEP and addressed the following: (1) What is the predator's (*A. meleagris*) density, feeding behavior, and predation pressure on branching pocilloporid colonies (coral prey)?, (2) What is the effect of puffer predation on the growth of coral colonies?, (3) What is the tolerance of coral colonies to injuries inflicted by the pufferfish?, and (4) What is the rate of destruction by puffers in relation to the rate of annual carbonate production by pocilloporid corals?.

Methods

Study site

This study was conducted from April to December 2012 at La Azufrada reef, a fringing coral reef located at Gorgona Island (2°58′27″N and 78°11′13″W) off the Colombian coast in the TEP region. La Azufrada reef is a shallow (4–12 m depth at high tide) calcareous framework extending over 9.4 ha, dominated by branching coral species of the genus *Pocillopora* (*P. capitata*, *P. damicornis*, *P. eydouxi*). Despite having relatively high coral cover (~40–75 %; Zapata et al. 2010), La Azufrada reef (like other reefs of the TEP) occurs within suboptimal and highly variable environmental conditions, such as an intense rainfall regime (resulting in frequent low salinity episodes and high water turbidity), sporadic extreme low tides, that cause subaerial exposure of corals and temperature alterations by ENSO events (Zapata and Vargas-Ángel 2003). During the time of the study, however, mean water temperature remained stable (27.7 ± 0.4 °C; mean ± SD) and oceanographic conditions corresponded to neutral/weak El Niño conditions (NOAA 2013).

Based on substrate cover composition and reef topography, three main reef zones can be distinguished at La Azufrada reef: (1) the back reef, which comprises a soft sediment bottom with many unattached coral colonies; (2) the reef flat, which is the largest reef zone and has a relatively thick (>8 m; Glynn et al. 1982) carbonate matrix where *Pocillopora* colonies are found both individually and in patches; and (3) the reef crest, where *Pocillopora* spp. form a dense, continuous stand that can cover up to 80–90 % of the substrate. As a result of the large variability in the abundance and distribution of pocilloporid colonies (preferred prey species for *A. meleagris*; Glynn et al. 1982; Guzmán and López 1991) throughout the coral reef, we considered observations from each of the three reef zones separately to account for variability in the pufferfish population and corallivore pressure.

Prey abundance (*Pocillopora* spp.)

Pocilloporid coral cover was estimated using 1 m^2 quadrats divided into 100 sub-quadrats of 10 × 10 cm^2. In each

zone, 30 quadrats were haphazardly placed and pocilloporid percentage cover was calculated for each as the number of sub-quadrats that were occupied by *Pocillopora* spp. (1 sub-quadrat = 1 % of cover). Differences in mean percentage of pocilloporid cover (arcsine-transformed) among reef zones (back reef, reef flat, and reef crest) were assessed using a resampling one-way ANOVA as the data did not satisfy the assumptions of normality and homoscedasticity required by standard parametric procedures. Subsequently a two-sample resampling test with Holm's correction ($\alpha = 0.05$) was used as a post hoc analysis to detect which means were significantly different from each other. The software Rundom Pro version 3.14 (Jadwiszczak 2009) was used to perform all the resampling analyses in this study.

Predator abundance (*Arothron meleagris*)

To estimate the population density of the puffer *A. meleagris*, we conducted 150 visual surveys along 50 × 5 m belt transects ($N = 50$ transects per reef zone). On each of the three reef zones, half of the surveys were conducted in the morning (09:22–11:55 h) and half in the afternoon (12:10–17:40 h) in order to obtain a uniform distribution of observations during the activity hours of *A. meleagris* (from ~6:00 h to ~18:30 h; personal observation). The number and the total length (TL; visually estimated to the nearest centimeter) of all puffers seen within transects were recorded. Puffer densities (ind ha^{-1}) were compared across reef zones (back reef, reef flat, and reef crest) through a resampling one-way ANOVA and a two-sample resampling test with Holm's correction ($\alpha = 0.05$). As data from fish sizes (TL) satisfied the assumptions normality and homoscedasticity of parametric tests, they were compared between the reef zones through a one-way ANOVA and a Tukey's multiple comparison test ($\alpha = 0.05$) as a post hoc test. We used the software STATISTICA version 8 (StatSoft Inc. 2007) to perform all parametric analyses in this study.

Standing bite density

All open lesions (exposed coral skeleton) on *Pocillopora* branch tips were considered to be bites made by the puffer *A. meleagris*, since preliminary field observations indicated that all conspicuous scars on coral branch tips were inflicted by *A. meleagris* (Fig. 1). No other fish species at Gorgona Island feeds in the same way, except for the congeneric *A. hispidus*, which is less abundant (23.7 ± 35.3 ind ha^{-1}, mean ± SD), occurs on deeper or sandy areas, and is rarely seen feeding on corals. Preliminary observations on the regeneration rate of coral tissue at La Azufrada reef indicated that superficial injuries heal in approximately 2 weeks and deeper lesions take about 2 months to recover. Consequently, we assumed that the standing density of feeding scars recorded in this study was the result of the puffers' feeding activity during the past few weeks.

The standing density of fish bite marks on *Pocillopora* colonies (bites m^{-2}) was estimated ($N = 90$ for the entire reef) with the 1 m^2 quadrats previously used to calculate coral cover (see "Prey abundance" section). Each time the large quadrat was haphazardly placed over the substrate, and bite marks were counted within those 10 × 10 cm sub-quadrats along the diagonal that had living coral (between three and ten sub-quadrats). Counts within the sub-quadrats in each quadrat were averaged and used as a replicate estimate of the standing bite density ($N = 30$ per reef zone). Standing bite density data were Log_{10} transformed to satisfy assumptions of normality and homoscedasticity. Differences in standing bite densities between reef zones (back reef, reef flat, and reef crest) were examined using a one-way ANOVA and a Tukey's multiple comparison test as a post hoc analysis ($\alpha = 0.05$). The spatial distribution of bite scars within each reef zone was analyzed by examining the fit to a Poisson distribution of the frequency distribution of the mean number of bites (rounded to the nearest integer) in the 10 x 10 cm^2 sub-quadrats. Lack of fit was determined using the Index of Dispersion (ID or variance to mean ratio), which follows a chi-square distribution with $N - 1$ degrees of freedom (Ludwig and Reynolds 1988).

Feeding rate

To determine the feeding rate of *A. meleagris*, we followed 120 haphazardly selected puffers ($N = 40$ puffers per reef zone) for ten min each while recording the number of bites taken from the substrate. The behavior and size of the fish were recorded, as well as the substrate type/species from which bites were taken. Fish feeding behavior was classified as active (≥1 bites 10 min^{-1}) or inactive (0 bites 10 min^{-1}). Half of the fish observations on each reef zone were conducted during the morning (08:40–11:40 h) and the other half during the afternoon (14:40–17:30 h). Puffer feeding rates (bites min^{-1}) were compared among reef zones (back reef, reef flat, and reef crest), among periods of the day (morning, 08:40–11:40 h and afternoon, 14:40–17:30 h) and among size classes of puffers (small: ≤17 cm TL, medium: 18–24 cm TL, and large: ≥25 cm TL) using resampling one-way ANOVAs as the data did not satisfy the assumptions of standard parametric tests.

Tolerance to corallivory experiment

The tolerance of pocilloporid colonies to different levels of damage caused by simulated pufferfish biting was

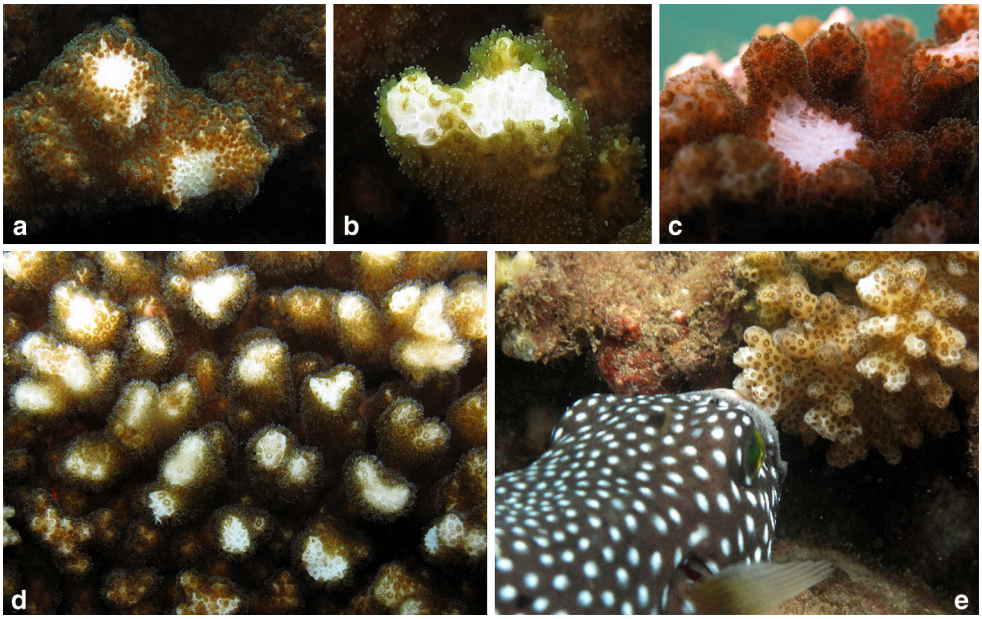

Fig. 1 Bite marks by the pufferfish *Arothron meleagris* on *Pocillopora* spp. colonies: **a** scrapes to the coral tissue, **b**, **c** excavating bites where coral tissue and skeleton have been removed, **d** pocilloporid colony exhibiting pufferfish bite marks, **e** *A. meleagris* feeding on *Pocillopora* spp.

determined through a field experiment conducted in the reef crest of La Azufrada. We monitored the tissue regeneration and growth of 40 small colonies of *Pocillopora* spp. (75–325 g) after exposure to four experimental treatments ($N = 10$ colonies per treatment) that differed in the percentage of branch tips per colony that was artificially damaged by simulated puffer bites: 0 % (control), 25, 50, and 75 % of branch tips scraped/tissue removed. Based on preliminary field observations, these damage treatments were designed to mimic the wide range of standing bite densities observed on pocilloporid colonies (47.7 ± 23.0 % of coral branches assessed had bite marks, mean ± SD, $N = 3{,}263$ branches) and to resemble the natural injuries inflicted by puffers (Fig. 1). Measurements with calipers (to the nearest 0.1 mm) of bite marks made by puffers on *Pocillopora* spp. revealed that bites were small (area = 0.47 ± 0.38 cm^2; mean ± SD; $N = 362$ bites) and that most often puffers inflicted only shallow injuries (mean bite depth = 0.19 ± 0.08 mm, $N = 50$ bites) on branch tips (although they also can occasionally break weak branches). In this experiment, artificial injuries made with pliers were similar in area to those measured in the field and although they were deeper (between 0.1 mm and 0.5 cm depth) than natural ones, we haphazardly made a large proportion of shallow scars (≤0.3 mm) to resemble the natural pattern. Considering that many puffer bite marks are superficial injuries that do not represent a significant removal of coral skeleton (Fig. 1a), the aim of our treatments was to injure the branch tips (resembling natural bite marks) in different densities, but not necessarily to remove specific weights of coral tissue/skeleton.

All colonies employed in this experiment were haphazardly collected from the reef and kept in tanks with fresh seawater, while they were being tagged, treated (according to the four experimental groups), and weighed (to the nearest 0.01 g). In <2 h from the time of collection, all colonies were taken to the reef and placed inside ten mesh cages (35 × 20 × 25 cm; 1.5 cm mesh size), so that each cage contained one colony from each treatment. Cages had previously been installed on the reef crest and were monitored periodically throughout the duration of the experiment to remove any fouling organisms and algae from the mesh. At the end of the experiment (122 d later), each colony was weighed for a second time. To assess pocilloporid tolerance to bite injuries and measure the degree to which coral fitness (in terms of growth) was affected, we compared the mean change in colony weight (g) among the four experimental groups (0, 25, 50, and 75 % of the branch tips injured by simulated puffer feeding) with a one-way ANCOVA using the initial colony weight as a covariate.

Predator exclusion experiment

The effect of puffer predation on the growth (measured as linear extension and weight change) of *Pocillopora* spp. was assessed in situ by exposing 30 nubbins, each ~4 cm long, to three experimental treatments ($N = 10$ per treatment): (1) predation treatment in which nubbins were kept

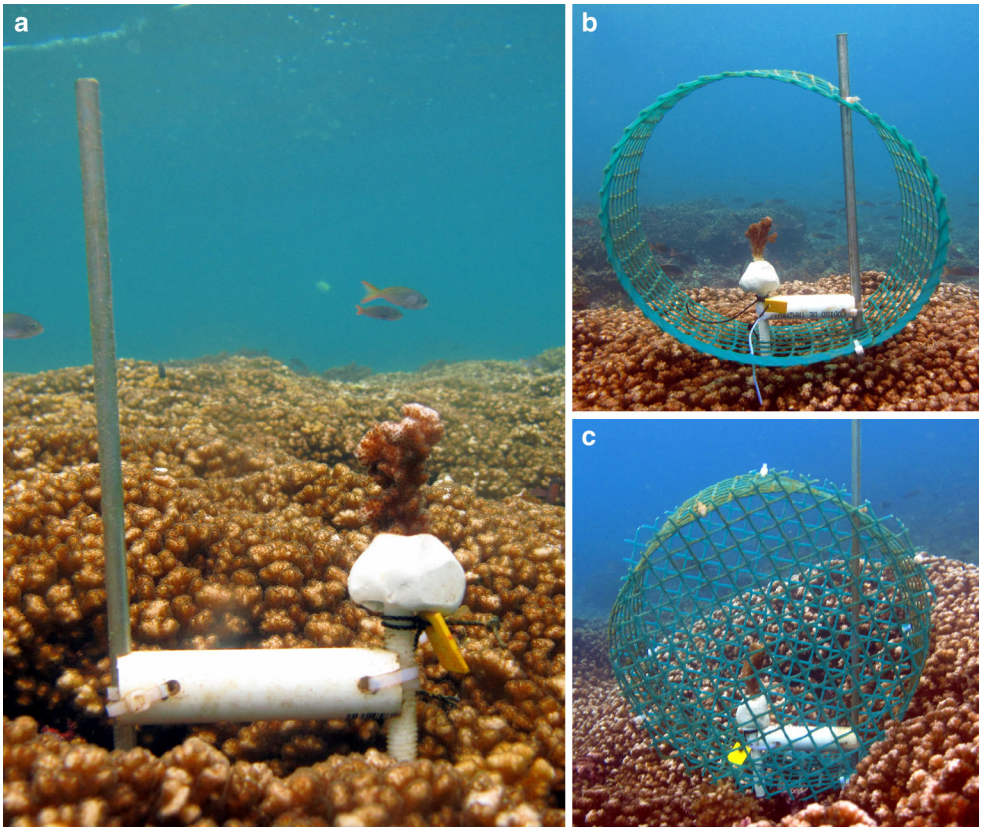

Fig. 2 Treatments considered in the Predator exclusion experiment where pocilloporid nubbins were individually exposed or protected from fish corallivores: **a** Predation treatment (no cage), **b** cage control treatment (half-cage), and **c** Predator exclusion treatment (full cage). Half-cages were employed to allow fish predation while controlling for the possible effects of the mesh on the nubbins

in the open, exposed to corallivory; (2) predator exclusion treatment in which mesh cages (25 × 25 × 11.5 cm; 1.5 cm mesh size) protected nubbins from fish corallivores; and (3) cage control treatment in which nubbins were kept within half-open cages (mesh cages lacking two sides) and therefore exposed to corallivory; this controlled possible effects of the full cage on the nubbins (Fig. 2). To conduct this experiment, haphazardly collected nubbins were individually glued (with epoxy cement) to plastic screws (Fig. 2), tagged, measured with calipers (lengthwise; from the base of the epoxy to the tip of the nubbin) to the nearest 0.01 cm, and weighed on a scale (including the screw and the tag) to the nearest 0.01 g. Afterward, they were placed on the reef flat at 5-m intervals, where the three treatments were systematically assigned to them in sequence. After 183 d, the nubbins were collected, measured, weighed, and examined closely for bite marks. Annotations were made on the number of the fish bites that each nubbin presented.

Changes in the length and weight of coral nubbins were normalized to surface area (cm cm^{-2} yr^{-1} and g cm^{-2} yr^{-1}) using the wax-dipping technique (Stimson and Kinzie 1991) to account for any differences in the size or surface area of the nubbins that might affect growth. To compare the normalized linear extension rate (cm cm^{-2} yr^{-1}) among the three treatments (predation treatment, predator exclusion treatment, and cage control treatment), we used a one-way ANOVA followed by a Tukey post hoc analysis. In contrast, as data from the normalized weight gained (g cm^{-2} yr^{-1}) did not satisfy the assumptions of parametric tests, differences between treatments were analyzed through a resampling one-way ANOVA. Additionally, because treatment-related changes in nubbin morphology (colony weight/length ratio) were apparent, we ran a parametric one-way ANOVA and Tukey's post hoc test to compare, across the three treatments, the average ratio of the normalized weight gain to the linear extension of nubbins (Δ weight/Δ length ratio).

The effect of puffers on the pocilloporid carbonate budget

We considered the effect of puffers on the pocilloporid standing crop as the percentage of the carbonate produced by pocilloporids that is removed by puffers. The carbonate production (kg m^{-2} yr^{-1}) of *Pocillopora* spp. was calculated as follows:

$$\text{Pocillopora Carbonate Production} = \text{Coral Cover } (\%) \times \text{Growth Rate}\left(\text{cm yr}^{-1}\right) \times \text{Colony Density}\left(\text{g cm}^{-3}\right)$$

where coral cover and growth rates were obtained as described previously (see the "Prey abundance" section for coral cover and the "Predator exclusion experiment" section for growth rate). Colony density was employed instead of skeletal density, as the latter should only be employed to calculate the carbonate production of massive coral colonies (where the density and volume of a colony are constant per unit of coral cover; Perry et al. 2012). Because ramose pocilloporid species have spaces between branches that do not contribute to carbonate production (discussed by Heiss 1995), we used the density of pocilloporid colonies and subtracted the volume of empty space to approximate carbonate production. Colony density (g cm^{-3}) was determined for five *Pocillopora* colonies that were individually wrapped with plastic film, weighed (g), and submerged in a container with water to estimate their volume (cm^3). Wrapping the entire colony in plastic film served to include the empty volume between branches in the estimation of the colony volume and thus colony density (Adams et al. 1985; Vytopil and Willis 2001).

On the other hand, the rate of Puffer Destruction (kg m^{-2} yr^{-1}) was obtained as:

$$\text{Puffer destruction} = \text{population density}\left(\text{ind ha}^{-1}\right) \times \text{destruction rate per fish}\left(\text{g min}^{-1}\right) \times \text{feeding correction factor.}$$

Data on puffer density were obtained from the fish surveys made in this study (see "Predator abundance" section), whereas the destruction rate per fish was calculated following fish bioerosion studies (e.g., Bellwood 1995; Bruggemann et al. 1996), in which the fish feeding rate (bites min^{-1}; see "Feeding rate" section) is multiplied by the mass of coral removed per bite (g $bite^{-1}$). To estimate the mean weight of coral removed per bite (g $bite^{-1}$), we followed the method used by Bellwood (1995) in which the volume of a bite mark (calculated by its length, width, and depth) is combined with the substratum density. Although the depth of bite marks is difficult to measure in branching pocilloporid species (original branch tip length is unknown), we did not deem appropriate the approach used by Glynn et al. (1972) as the puffers significantly change their feeding behavior when enclosed in mesh cages (personal observation). For our calculations, we used the bite mark measures taken for the "Tolerance to corallivory experiment" and the mean skeletal density of 30 *Pocillopora* spp. branches collected in the field. To obtain the density estimates (g cm^{-3}), branches were individually weighed with a digital scale to the nearest 0.001 g and volume was calculated as the water displaced by each branch in a 50-ml graduated cylinder (±0.1 ml). Finally, the feeding correction factor in the formula reflects the proportion of puffers actively feeding at a given time and was calculated from our field observations (see "Feeding rate" section).

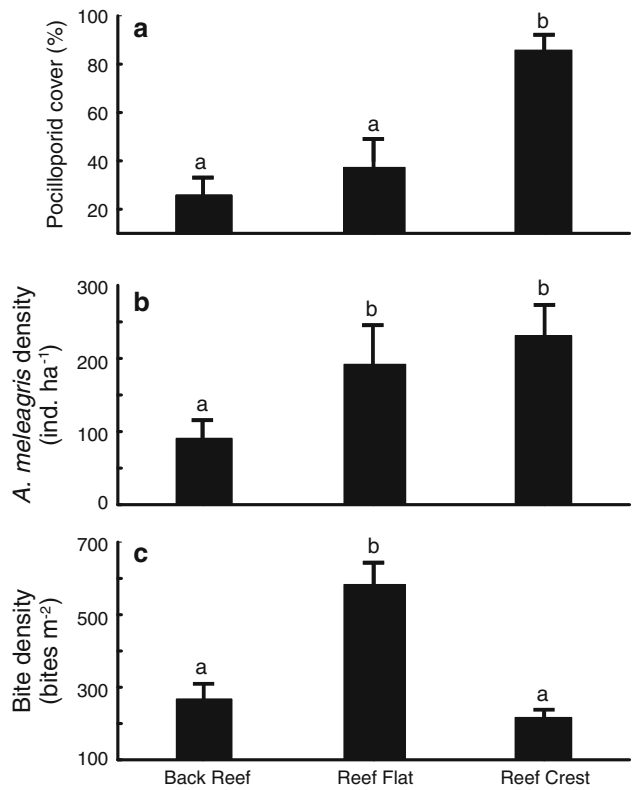

Fig. 3 Mean (+95 % CI) **a** pocilloporid coral cover (%), **b** *Arothron meleagris* abundance (ind ha^{-1}), and **c** standing bite density (bites m^{-2}) across three zones of La Azufrada coral reef, Gorgona Island. *Bars* with the *same lowercase letter* did not differ significantly: **a**, **b** two-sample resampling tests with Holm's correction ($\alpha = 0.05$), **c** Tukey's multiple comparison test ($\alpha = 0.05$)

Results

Prey abundance (*Pocillopora* spp.)

The average cover of pocilloporid colonies at La Azufrada reef was 49.6 ± 37 % (mean ± SD, $N = 90$). *Pocillopora* spp. cover varied significantly across reef zones ($F_{2,87} = 43.3$, $P < 0.001$; Fig. 3a), with the highest cover recorded on the reef crest, where a dense and continuous stand of interlinked pocilloporid colonies occupied 85.8 ± 18.7 % ($N = 30$) of the substrate. Both the back reef and the reef flat had a coral cover of <40 % as the back reef was dominated by sand and rubble, and the reef flat was characterized by large areas of calcareous matrix

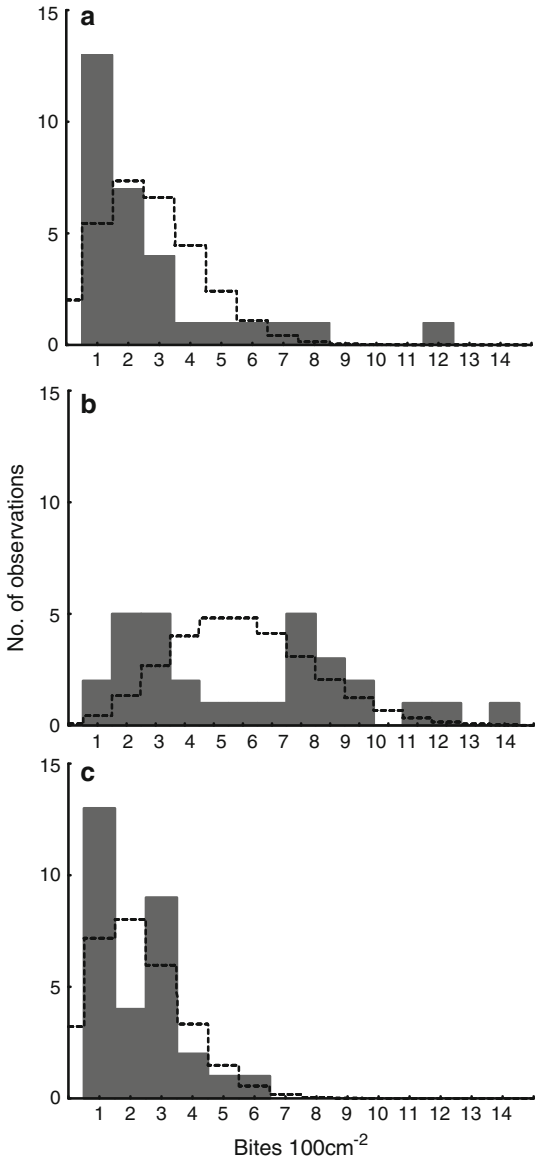

Fig. 4 Observed frequency distribution of standing bite scars within 100 cm^2 quadrats on each of three reef zones at La Azufrada coral reef: **a** back reef, **b** reef flat, and **c** reef crest. *Dash line histograms* correspond to the expected Poisson distribution under the null hypothesis of a random spatial distribution of bite scars

(composed of coral rubble tightly cemented by encrusting calcareous algae) and scattered coral patches.

Predator abundance (*Arothron meleagris*)

The puffer *A. meleagris* had an average density of 171.2 ± 166.9 ind ha^{-1} (mean ± SD, $N = 150$) and an estimated total population size of 1,606 individuals (on a 9.4 ha reef). Significant differences in puffer abundance were found among reef zones ($F_{2,147} = 10.7$, $P < 0.001$; Fig. 3b), with the reef crest (231.2 ± 160.7 ind ha^{1}, $N = 50$) and the reef flat (192 ± 199.7 ind ha^{-1}, $N = 50$) being the zones with the highest puffer density. The abundance of puffers increased toward the outer edge of the reef, where puffers usually aggregate in groups of up to 15 individuals and hide in caves within the pocilloporid framework (Glynn et al. 1982). Puffer size at La Azufrada ranged between 10 and 30 cm TL and had a mean TL of 20.5 ± 3.9 cm ($N = 642$). Puffers from the back reef were significantly smaller than the ones from the reef flat and the reef crest ($F_{2,612} = 5.3$, $P < 0.05$).

Standing bite density

At La Azufrada reef, the average puffer-inflicted standing bite density on *Pocillopora* spp. colonies was 366.2 ± 329.7 bites m^{-2} (mean ± SD, $N = 90$). However, standing bite density differed significantly among the three reef zones ($F_{2,87} = 15.5$; $P < 0.001$; Fig. 3c). The reef flat, which occupies the most area (58 %) of La Azufrada reef, had a higher standing bite density (608.6 ± 383.7 bites m^{-2}, $N = 30$) than the other reef zones (<300 bites m^{-2}), indicating a higher feeding pressure by the puffers on the pocilloporid patches occurring on the reef flat. The spatial distribution of bite scars was aggregated on the back reef and the reef flat (back reef: ID = 2.45, $P < 0.005$, $df = 29$; reef flat: ID = 2.47, $P < 0.05$, $df = 29$), but uniform on the reef crest (ID = 0.82, $P < 0.005$, $df = 29$; Fig. 4) where coral cover was the highest.

Feeding rate

The majority of *A. meleagris* bites was made on live *Pocillopora* spp. colonies (86.6 % of all bites), although *Psammocora stellata* (especially in the back reef, where it is more abundant), coral rubble, and other invertebrates (e.g., sponges) were also bitten by the fish. Sixty percent of the 150 pufferfish we followed were actively feeding, so we used this value to estimate the rate of coral consumption by puffers. During the feeding activity, the puffers' average feeding rate on *Pocillopora* was 1.8 ± 1.4 bites min^{-1} (mean ± SD, $N = 72$) with no significant differences among fish size classes ($F_{2,69} = 0.8$, $P > 0.3$), reef zones ($F_{2,69} = 0.98$, $P > 0.3$), or periods of the day, ($F_{1,70} = 0.13$, $P > 0.9$).

Tolerance to corallivory experiment

Uninjured control colonies of *Pocillopora* spp. were slightly heavier than injured colonies at the end of the experiment; however, their change in weight was not statistically different from the weight changes of colonies in any of the damage treatments ($F_{3,31} = 1.12$, $P > 0.1$;

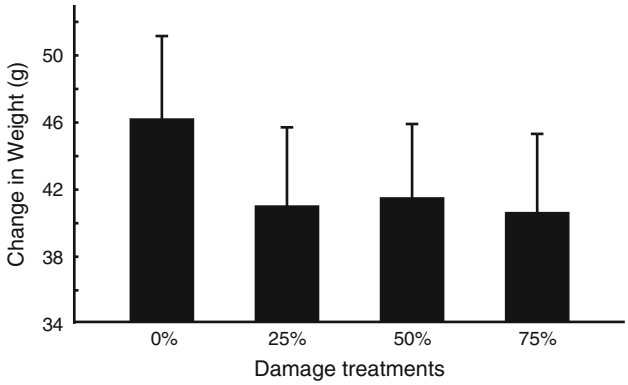

Fig. 5 Adjusted mean (+95 % CI) of the change in weight attained by pocilloporid colonies exposed to different levels of simulated bite-inflicted damage (control = 0 % or no damage, and 25, 50, or 75 % of colony branch tips damaged). Means of change in weight were adjusted by ANCOVA to the mean initial weight of colonies

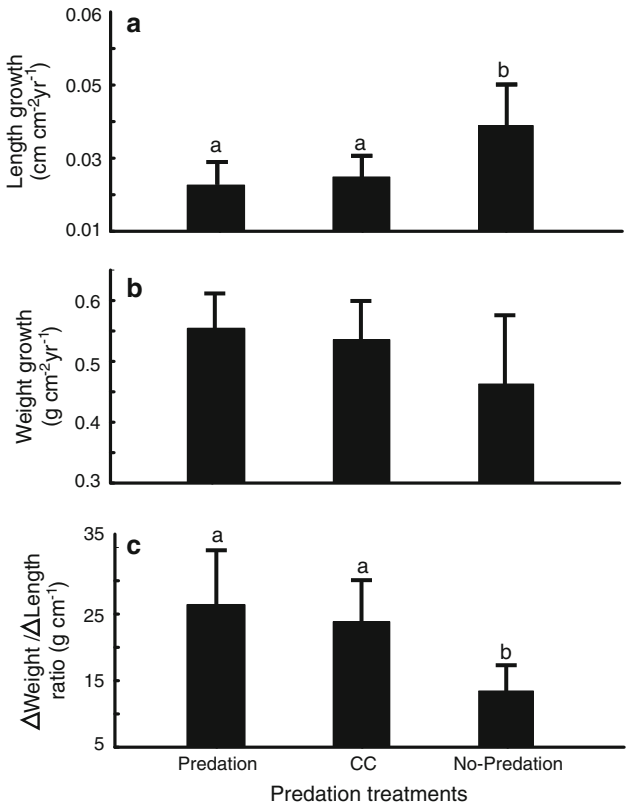

Fig. 6 Mean (+95 % CI) growth in terms of **a** normalized linear extension (cm cm^{-2} yr^{-1}) and **b** normalized weight (g cm^{-2} yr^{-1}) of *Pocillopora* spp. nubbins in each of three predation treatments (Predation, cage control = CC, and Predation exclusion). **c** Average ratio between the normalized weight gained and the increased linear extension of nubbins (normalized) in each treatment. *Bars* with the *same lowercase letter* did not differ significantly: **a**, **c** Tukey multiple comparison test ($\alpha = 0.05$), **b** two-sample resampling tests with Holm's correction ($\alpha = 0.05$)

Fig. 5). Growth rate of pocilloporid colonies seemed not to be affected significantly by the levels of injury we examined (25, 50, or 75 % of the branch tips injured by

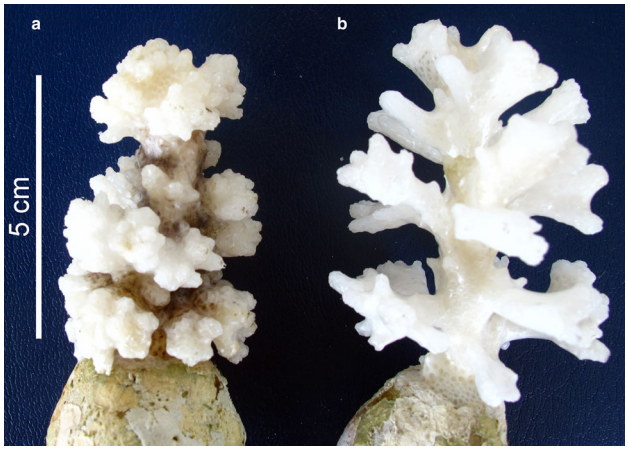

Fig. 7 Representative corallum morphologies of pocilloporid nubbins **a** exposed and **b** protected from fish corallivory in the Predator exclusion experiment carried out at La Azufrada coral reef, Gorgona Island, Colombia

simulated puffer feeding). All injuries had healed successfully after 2 months with no apparent negative effects on the colony branches. Thus, pocilloporid colonies appear to tolerate bitelike injuries on up to 75 % of their branches.

Predator exclusion experiment

Pocilloporid nubbins exposed to corallivory (predator and cage control treatments) had an average of 15.2 ± 9.3 fish bites $nubbin^{-1}$ (mean ± SD, $N = 20$) and a significantly lower normalized linear growth than the protected nubbins ($F_{2,26} = 5.1$, $P < 0.05$; Fig. 6a). Remarkably, nubbins in all treatment groups had a similar normalized weight gain ($F_{2,26} = 1.7$, $P = 0.2$; Fig. 6b). Nubbins in the half-cage treatment did not differ in linear growth or weight gain from those in the predator (uncaged) treatment, so it was assumed that the mesh did not affect growth of the fully caged nubbins (predator exclusion treatment).

Results from the Predator exclusion experiment provided evidence that the presence/absence of predation influenced the morphology (colony weight/length ratio) and normalized growth rates (Δ weight/Δ length ratio) of pocilloporid nubbins: Corals exposed to corallivory were usually thick and short and had a greater rate of increase in weight for a given rate of linear extension than corals protected from fish predation ($F_{2,26} = 6.2$, $P < 0.01$; Figs. 6c, 7). Nubbins in the predator exclusion treatment showed the opposite morphology (long thin branches).

Effect of puffers on the pocilloporid carbonate budget

The puffer *A. meleagris* removed 15.8 % of the carbonate produced by the pocilloporid stands at La Azufrada reef, assuming a mean carbonate production rate of 5.5 kg

Table 1 Carbonate budget estimated for La Azufrada coral reef, considering only the production of *Pocillopora* spp. and the destruction by the puffer *Arothron meleagris*

Pocillopora spp. production				Destruction by *A. meleagris*			
Growth rate (cm yr^{-1})	Colony density (g cm^{-3})	Cover (%)	Production (kg m^{-2} yr^{-1})	Density (ind ha^{-1})	Destruction per fish (g d^{-1})	CF	Destruction (kg m^{-2} yr^{-1})
2.70	0.42	49	5.56	171.2	233.28	0.6	0.87

The destruction rate per fish was calculated taking into account a carbonate removal of 0.18 ± 0.14 g $bite^{-1}$ (mean ± SD), a feeding rate of 1.8 ± 1.4 bites min^{-1} and 12 h of feeding activity per day (based on field observations). CF is a correction factor representing the proportion of the pufferfish population actively feeding at a given time

Table 2 Abundance of the puffer *Arothron meleagris* on pocilloporid reefs within the Tropical Eastern Pacific

Site	Date of survey	Reef size (ha)	Coral cover (%)	*A. meleagris* (ind ha^{-1})	References
Colombia					
Azufrada reef	1979	15	36.5	12	Glynn et al. (1982)
Azufrada reef	1987/1988	15	48.5	34	Guzmán and Robertson (1989)
Playa Blanca reef	1989	8	32.6	21.9	Guzmán and López (1991)
Azufrada reef	1989	15	50.2	25.8	Guzmán and López (1991)
Azufrada reef	1993	10	–	80	Zapata and Morales (1997)
Azufrada reef	2012	9.4	49.6	171.2	This study
Mexico					
Cabo Pulmo reef	1991/1992	220	30	39	Reyes-Bonilla and Calderon-Aguilera (1999)
Playa Mora reef	2002/2004	2.5	–	43.7	Galván-Villa et al. (2011)
Panama					
Señora Is. reef	1971	0.8	75	40	Glynn et al. (1972)
Uva Is. reef	1981/1984	2.4	45	55	Glynn (1985)
Uva Is. reef	1986/1988	2.4	28.2	55	Guzmán and Robertson (1989)
Secas Is. reef	1986/1988	7.6	13.8	27	Guzmán and Robertson (1989)
Señora Is. reef	1987	0.8	49.5	0	Guzmán and Robertson (1989)

The location, reef size, percentage of live cover, date of survey and sources are shown. Modified from Guzmán and Robertson (1989)

$CaCO_3$ m^{-2} yr^{-1} and a puffers' destruction rate of 0.87 kg $CaCO_3$ m^{-2} yr^{-1} (Table 1).

Discussion

This study determined that a large proportion of La Azufrada pocilloporid reef is under intense corallivore pressure from the puffer *A. meleagris*. Coral colonies on the reef flat, the largest reef zone at La Azufrada reef (58 % of the entire reef's area), are consumed intensely by puffers. The mean standing bite density observed on pocilloporid colonies (~600 bites m^{-2}) was greater than the median of the mean bite densities reported for several massive and branching coral species around the world (4–952 bites m^{-2}; Rotjan and Lewis 2006; Jayewardene et al. 2009; Bonaldo and Bellwood 2011; Roff et al. 2011). Although corallivore pressure is influenced by several factors (e.g., predator feeding rates, abundance, bite size), the impact recorded here is mainly due to the high density of *A. meleagris* (the highest ever reported for this species on any coral reef within the TEP; Table 2), as estimates of its feeding rate (1.8 bites min^{-1}) and carbonate removal per bite (0.18 g $bite^{-1}$) are similar to or lower than estimates for excavating parrotfishes (*Chlorurus gibbus*, *C. sordidus*, and *Sparisoma viride*; 6.7–18.0 bites min^{-1} and 0.05–2.7 g $bite^{-1}$; Bellwood 1995; Reyes-Nivia et al. 2004). As expected, high densities of *A. meleagris* cause the greatest standing bite density when coral cover is relatively low and patchy. Although the reef crest and reef flat had similar puffer densities, the pocilloporid colonies on the reef flat exhibited greater bite densities as predators concentrated their feeding on the limited coral patches (Jayewardene et al. 2009; Shantz et al. 2011). The spatial

distribution of bite scars further supports this idea, since bites were aggregated on the reef flat, but uniformly distributed on the reef crest where coral cover was greatest. Together, this body of information suggests that *A. meleagris* exert a substantial impact to the distribution and growth of corals in La Azufrada.

While pufferfish predation affects the linear growth and morphometrics of *Pocillopora* spp. colonies, these colonies have the capacity to tolerate and recover from the high intensity levels of injury. As was expected, the constant pruning and trimming of branch tips by fish corallivores (Glynn et al. 1982) meant that nubbins exposed to predation increased their length at a lower rate than those protected from predation. However, similar to results reported for *Porites porites* in the Caribbean (Fig. 5b in Miller and Hay 1998), *Pocillopora* spp. appear to tolerate partial predation as the colony carbonate production (in terms of weight gain) and recovery were not negatively affected, despite corallivory causing $\sim$15 bites nubbin^{-1} or damage on up to 75 % of all branch tips in a colony. In contrast to massive corals, in which injuries are highly deleterious because their growth is relatively slow and recovery requires the deposition of new skeleton within the lesion (compensatory growth; sensu Henry and Hart 2005), pocilloporid corals can tolerate injuries because their healing process is not regenerative *per se*, but occurs through stimulation of the apical points of growth (branch tips) or reoriented growth (Lenihan and Edmunds 2010). Our results support the "reoriented growth hypothesis" as the morphology of pocilloporid nubbins exposed to predation changed significantly as shown by the redistribution of growth (Δ weight/Δ length ratio).

The propensity of *Pocillopora* spp. to change their phenotype in response to environmental conditions (e.g., sediment, light, or depth; Veron 1995; Todd 2008) means that an important biological stressor such as corallivory is very likely to change morphometric proportions of pocilloporid colonies. Predator-induced phenotypic variation has been reported in several clonal–marine organisms (Gaulin et al. 1986; West 1997; Hill and Hill 2002). For instance, polyp withdrawal and changes in nematocyst density have been observed as responses to predation in hard corals (Gochfeld 2004; Rotjan and Dimond 2010). However, a recent review of morphological plasticity in scleractinian corals (Todd 2008) found no previous evidence of morphological skeletal change induced by fish predation in *Pocillopora* spp. The thick and short shape of nubbins that were exposed to corallivory (Fig. 7) results from the accumulation of carbonate through the thickening of the branches given their reduced linear growth (as the branch tips are constantly being cropped).

Despite intense feeding, puffer predation is not having a major effect on the carbonate production of individual pocilloporid colonies (measured as weight gain) or on the carbonate budget of the entire pocilloporid reef. As in other pocilloporid reefs within the TEP (Glynn et al. 1972; Reyes-Bonilla and Calderon-Aguilera 1999), this study has shown that the destruction rate of *A. meleagris* (0.87 kg m^{-2} yr^{-1}) is not that high (<16 %) compared with the annual pocilloporid carbonate production at La Azufrada reef (5.5 kg m^{-2} yr^{-1}). However, it should be noted that our estimates of puffer impact and carbonate production rates were different from those reported in other studies, probably due to methodological differences. For instance, we estimated carbonate production rates using a colony density of 0.42 g cm^{-3}, however, if we had used data from Eakin (1996) our carbonate production would have been lower (2.2 kg m^{-2} yr^{-1}). In terms of the impact by *A. meleagris*, Glynn et al. (1972) and Reyes-Bonilla and Calderon-Aguilera (1999) used daily ingestion rates from puffers held in aquaria to estimate the daily mass removed from colonies (16–20 g m^{-2} d^{-1}). If we use such estimates (which may be an underestimate due to changes in the feeding behavior of puffers in aquaria; personal observation), the puffers' destruction rate at La Azufrada would be eight times lower (0.1 kg m^{-2} yr^{-1}). In a scenario where the carbonate production is the lowest (2.2 kg m^{-2} yr^{-1}; using the colony density from Eakin 1996) and the puffers' destruction is the highest (using all of our data), the annual pocilloporid production removed by *A. meleagris* predation would correspond approximately to 39.5 %. Such an increase in the estimate suggests that where corallivory is intense, a relatively high carbonate production appears to be essential for a significant reef accretion and persistence.

In conclusion, despite of the fact that high corallivory limits their linear growth, *Pocillopora* colonies can withstand partial predation. Colonies can easily recover from injuries and maintain their carbonate production rate through the thickening of branches. Under the ecological conditions in this study, pufferfish predation does not prevent reef accretion. Predation by *A. meleagris* may only pose a risk to the *Pocillopora* colonies or the reef framework when coincident with major coral mortality or the presence of acute environmental stressors (e.g., aerial exposure, ENSO events; Zapata and Vargas-Ángel 2003). At these times, predators will tend to concentrate on the remaining live coral colonies (causing increased bite density) and corals will be less physiologically capable of responding to the impact of tissue and skeletal damage. A noteworthy finding of this work was that corallivory causes phenotypic changes in hard corals and consequently can be the source of irregular morphologies reported in these organisms (Jayewardene and Birkeland 2006). In the case of Pocilloporid corals, which are already known for their morphological variability and phenotypic plasticity, this issue exacerbates the difficulties inherent in taxonomic

studies (Veron 1995; Pinzón et al. 2013; Schmidt-Roach et al. 2014). Determining the extent to which predation affects morphological variability will be an important step in understanding phenotypic plasticity in this group. A more complete understanding of the importance of corallivory in pocilloporid reefs will involve assessing the effects of predation on sexual and asexual coral reproduction, including their relative contribution to population replenishment, and a complete carbonate budget of La Azufrada coral reef that includes other producers and bioeroders.

In contrast with other important processes such as herbivory, the role of corallivory on coral reefs has only recently received attention. While the importance of corallivory has perhaps been underestimated, it is now becoming apparent that corallivores can have diverse impacts on coral fitness, coral population and community structure, reef structure, and ecosystem function (Rotjan and Lewis 2008). Given the increasing levels of natural and anthropogenic disturbances on coral reefs, it is important to examine the full spectrum of impacts by corallivores on corals and coral reefs, and whether they contribute to decrease reef resilience. Our study provides one example from the TEP, where reefs are somewhat simple, low diversity systems, in which intense predation by a major corallivore does not seem to negatively affect reef resilience. Studies of corallivory in locations that differ in coral diversity and patterns of disturbance are required to fully understand the role of corallivory on coral reefs.

Acknowledgments Financial support for this study was provided by Universidad del Valle and the Administrative Department of Science, Technology and Innovation of Colombia (Colciencias) through a Young Researcher and Innovator Program award ("Virginia Gutiérrez de Pineda") to M.M. Palacios, a scholarship for doctoral studies awarded to C.G. Muñoz and a research Grant (1106-489-25135) awarded to F.A. Zapata. We are grateful to UAESPNN (Colombian National Parks) for providing logistical support at Gorgona Island and to members of the Coral Reef Ecology Research Group at Universidad del Valle, mainly Luis David Lizcano and Ana Lucía Castrillón, for their help during field work. We also thank D.P. Manzello and C.T. Perry for their comments regarding carbonate production by ramose corals. An earlier version of this manuscript was improved thanks to comments made by two anonymous reviewers and H. Sweatman.

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